

Krylov Solvability for Scalar Friedrichs Transport Operators: Stable Groups, Quasi-Analytic Data, and a Smooth Obstruction

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Status and scope

Candidate coefficient-wide strong partial resolution, likely valid, pending specialist review. Question 3 of Caruso and Michelangeli asks for an investigation of Krylov solvability for

$$-\frac{d}{dx} + 1 \quad \text{and} \quad -\frac{d}{dx} + c(x)$$

under suitable conditions on the datum and coefficient [1]. This packet treats the full scalar coefficient class singled out there: for the maximal closed realization and

$$c \in L^2_{\text{loc}}(\mathbb{R}), \quad 0 < \delta \leq \operatorname{Re} c(x) \leq M < \infty \quad \text{a.e.},$$

it proves Krylov solvability for every quasi-analytic datum, gives an exact weighted-derivative criterion for every smooth datum, and shows the positive class is dense. The constant-coefficient construction from the preceding version supplies a Schwartz datum for which solvability fails, so mere A -smoothness is insufficient.

Because the source question is an open-ended request to “investigate” rather than a precise universal conjecture, the packet is conservatively retained as **partial**. Mathematically, however, the coefficient side is no longer left open: the theorem covers every scalar coefficient satisfying the source’s bounded coercive-real-part hypothesis. What is not supplied is a simpler closed-form classification of all smooth data beyond the exact closure criterion.

Source question

The source question occurs in Section 4, PDF page 11. The crop includes the constant operator (4.4), the variable operator (4.5), and Question 3.

Friedrichs systems.

This has naturally brought attention to those joint pairs $(A_0, \tilde{A}_0)$ of abstract Friedrichs operators such that

$$(4.3) \quad A_0 + \tilde{A}_0 = \alpha \mathbb{1} \quad \alpha \in \mathbb{R}.$$

Consider for instance this very prototypical Friedrichs operator: the complex Hilbert space $\mathcal{H} = L^2(\mathbb{R})$, and

$$(4.4) \quad \text{dom } A := H^1(\mathbb{R}), \quad Af := -f' + f.$$

A is actually an unbounded Friedrichs operator, as it is closed and its adjoint acts as $\frac{d}{dx} + \mathbb{1}$ on $H^1(\mathbb{R})$, thereby implying that $A + A^* = 2\mathbb{1}$ is obviously bounded, self-adjoint, and coercive (see (3.3) above).

More generally, one may consider variants where A acts as

$$(4.5) \quad A = -\frac{d}{dx} + c(x) \\ \text{for } c : \mathbb{R} \rightarrow \mathbb{C} \text{ such that } c(x) + \overline{c(x)} \text{ is a} \\ \text{positive bounded function separated from zero}$$

(The second term in $-\frac{d}{dx} + c(x)$ indicates the multiplication by the function c .)

QUESTION 3: *To investigate the Krylov-solvability of the inverse problem $Af = g$ induced by (4.4) or (4.5) (or variants) under suitable conditions on g and c .*

It is not difficult to argue that the general arguments preceding Question 2, as well as the above-mentioned partial answers to Question 2, may be made applicable to Question 3.

For instance, we could show [15] that in the case (4.4) it suffices to assume further that the $\hat{g} \in C_c^\infty(\mathbb{R})$ (i.e., the Fourier transform of g is smooth and compactly supported) in order to deduce that A^*g is a *bounded* (and hence quasi-analytic) vector for A^*A , in the sense of Definition 2.6. This ensures that A^*g is in particular

The question asks: “To investigate the Krylov-solvability of the inverse problem $Af = g$ induced by (4.4) or (4.5) (or variants) under suitable conditions on g and c .” The paper then records $\hat{g} \in C_c^\infty(\mathbb{R})$ as a sufficient condition for (4.4).

Definitions

For a closed operator A on a Hilbert space H , put

$$C^\infty(A) = \bigcap_{n \geq 0} \text{dom}(A^n), \quad \mathcal{K}(A, g) = \text{span}\{g, Ag, A^2g, \dots\}.$$

The inverse problem $Af = g$ is *Krylov-solvable* when it has a solution in $\overline{\mathcal{K}(A, g)}$ [2, 3]. A nonzero $g \in C^\infty(A)$ is *quasi-analytic for A* when

$$\sum_{n=1}^{\infty} \|A^n g\|^{-1/n} = \infty. \quad (1)$$

This is the standard quasi-analytic-vector condition appearing in the Denjoy–Carleman/operator-theoretic correspondence [6].

Proof intuition

There are two changes of viewpoint. First, a general stable-group lemma says that if $-A$ generates a group whose positive orbit decays exponentially, then the inverse is the orbit integral $A^{-1}g = \int_0^\infty e^{-tA}g \, dt$. For a vector satisfying (1), every scalar orbit coefficient is quasi-analytic. Its entire Taylor jet at zero is determined by the Krylov vectors. Any vector orthogonal to the Krylov closure therefore annihilates the whole orbit, and hence annihilates its integral.

Second, the integrating factor e^{-C} , where $C(x) = \int_0^x c(s) ds$, does not give a bounded similarity on ordinary L^2 . It instead gives an exact *unitary* equivalence to differentiation on the weighted space $L^2(e^{2\operatorname{Re} C} dx)$. There the group is ordinary right translation, and coercivity of $\operatorname{Re} c$ is exactly what makes positive translations decay. This proves the coefficient-wide positive theorem and also turns arbitrary-data solvability into an exact weighted derivative-approximation question.

The positive condition is not cosmetic. At $c = 1$, Fourier transformation turns the problem into polynomial approximation in a datum-dependent measure. An affine lognormal moment construction produces a Schwartz datum with a nonzero polynomial-orthogonal witness that still pairs nontrivially with the resolvent. Its power norms grow at least like e^{n^2} , placing it strictly outside the quasi-analytic class.

A stable-group Krylov theorem

Theorem 1 (Stable group plus quasi-analyticity). *Let A be closed and densely defined on a Hilbert space H . Suppose $-A$ generates a strongly continuous group $T(t)$ and, for some $M_0, \delta > 0$,*

$$\|T(t)\| \leq M_0 e^{-\delta t} \quad (t \geq 0). \quad (2)$$

Then A is boundedly invertible. For every nonzero A -quasi-analytic vector g , the unique solution of $Af = g$ is Krylov-solvable, and

$$A^{-1}g = \int_0^\infty T(t)g dt \in \overline{\mathcal{K}(A, g)}. \quad (3)$$

Proof. The Bochner integral $R = \int_0^\infty T(t) dt$ converges in operator norm by (2). The standard generator identity

$$(-A) \int_0^N T(t)x dt = T(N)x - x$$

and closedness, followed by $N \rightarrow \infty$, give $ARx = x$ for all $x \in H$. For $x \in \operatorname{dom} A$, differentiating the orbit and integrating gives $RAx = x$. Hence $R = A^{-1}$.

Fix a quasi-analytic g , put $M_g = \overline{\mathcal{K}(A, g)}$, and let $0 \neq h \perp M_g$. Define on $\mathbb{R}$

$$\phi_h(t) = \langle T(t)g, h \rangle.$$

The smooth-vector theorem for groups gives

$$\phi_h^{(n)}(t) = \langle T(t)(-A)^n g, h \rangle.$$

On every compact interval I there is a constant C_I such that

$$\sup_{t \in I} |\phi_h^{(n)}(t)| \leq C_I \|A^n g\| \|h\| \quad (n \geq 0). \quad (4)$$

Multiplication of the derivative majorants by a fixed positive constant does not change divergence of their inverse-root sum. More explicitly, for $M_n = C_I \|A^n g\| \|h\|$ and $\beta_n = \inf_{k \geq n} M_k^{1/k}$, one has $\beta_n^{-1} \geq M_n^{-1/n}$; hence Carleman's criterion $\sum_n \beta_n^{-1} = \infty$ holds. The Denjoy–Carleman theorem therefore makes ϕ_h quasi-analytic on every compact interval [6]. Since

$$\phi_h^{(n)}(0) = \langle (-A)^n g, h \rangle = 0 \quad (n \geq 0),$$

quasi-analyticity gives $\phi_h \equiv 0$ on $\mathbb{R}$. Thus $T(t)g$ is orthogonal to $M_g^\perp$ and belongs to M_g for every t . Closedness of M_g and the norm-convergent integral in (3) imply $A^{-1}g \in M_g$. $\square$

The scalar variable-coefficient model

Let $c \in L^2_{\text{loc}}(\mathbb{R})$ satisfy

$$0 < \delta \leq \operatorname{Re} c(x) \leq M < \infty \quad \text{a.e.} \quad (5)$$

Set

$$C(x) = \int_0^x c(s) ds, \quad \omega(x) = e^{2\operatorname{Re} C(x)}, \quad H_\omega = L^2(\mathbb{R}, \omega(x) dx).$$

On H_ω define

$$Bq = -q', \quad \operatorname{dom} B = \{q \in H_\omega : q' \in H_\omega\},$$

where the derivative is distributional. Multiplication

$$U : H_\omega \rightarrow L^2(\mathbb{R}), \quad Uq = e^C q,$$

is unitary. Define $A_c = UBU^{-1}$.

Theorem 2 (Coefficient-wide resolution). *Under (5), A_c is the maximal closed realization of $-d/dx + c(x)$ and is boundedly invertible. If $c \in L^\infty(\mathbb{R})$, then $\operatorname{dom} A_c = H^1(\mathbb{R})$. Moreover:*

(i) *Every nonzero $g \in C^\infty(A_c)$ satisfying*

$$\sum_{n=1}^{\infty} \left\| \left(\frac{d}{dx} \right)^n (e^{-C} g) \right\|_{L^2(\omega dx)}^{-1/n} = \infty \quad (6)$$

has a Krylov solution.

(ii) *For arbitrary $g \in C^\infty(A_c)$, put $q = e^{-C} g$. The unique solution is*

$$f(x) = e^{C(x)} \int_x^\infty e^{-C(y)} g(y) dy, \quad (7)$$

and it is a Krylov solution if and only if

$$x \mapsto \int_x^\infty q(y) dy \quad \text{belongs to} \quad \overline{\operatorname{span}\{q, q', q'', \dots\}}^{H_\omega}. \quad (8)$$

(iii) *The A_c -analytic vectors, hence also the vectors satisfying (6), form a dense subspace of $L^2(\mathbb{R})$.*

Proof. The unitary identity $\|Uq\|_2 = \|q\|_{H_\omega}$ is immediate. Since $C' = c$ locally in the weak sense,

$$UBU^{-1}f = e^C(-(e^{-C}f)') = -f' + cf.$$

Thus the unitary definition gives precisely the maximal distributional realization of the displayed expression. When c is bounded, multiplication by c is bounded on L^2 , so its domain is $H^1(\mathbb{R})$.

The group generated by $-B$ is right translation,

$$S(t)q(x) = q(x+t).$$

Strong continuity follows first on compactly supported smooth functions and then by density, using the local comparability of translates of ω . For $t \geq 0$, the change of variable $y = x + t$ gives

$$\begin{aligned} \|S(t)q\|_{H_\omega}^2 &= \int_{\mathbb{R}} |q(y)|^2 \omega(y-t) dy \\ &= \int_{\mathbb{R}} |q(y)|^2 \omega(y) \exp\left(-2 \int_{y-t}^y \operatorname{Re} c(s) ds\right) dy \leq e^{-2\delta t} \|q\|_{H_\omega}^2. \end{aligned} \quad (9)$$

For negative time the analogous estimate is $\|S(t)\| \leq e^{M|t|}$, so S is a C_0 group. Conjugating by U gives

$$T_c(t)g(x) = \exp\left(-\int_0^t c(x+s) ds\right) g(x+t), \quad (10)$$

and (9) gives $\|T_c(t)\| \leq e^{-\delta t}$ for $t \geq 0$. Theorem 1 applies.

If $g \in C^\infty(A_c)$ and $q = U^{-1}g = e^{-C}g$, unitary equivalence gives

$$\|A_c^n g\|_2 = \|B^n q\|_{H_\omega} = \|q^{(n)}\|_{H_\omega}, \quad U^{-1}\overline{\mathcal{K}(A_c, g)} = \overline{\text{span}\{q, q', q'', \dots\}}^{H_\omega}. \quad (11)$$

Thus (6) is exactly (1), and part (i) follows. Also,

$$B^{-1}q = \int_0^\infty S(t)q dt,$$

which has the almost-everywhere representative $B^{-1}q(x) = \int_x^\infty q(y) dy$. This integral is absolutely convergent for each fixed x by Cauchy–Schwarz and the exponential lower growth of ω . Applying U gives (7); comparing it with (11) proves the exact criterion in part (ii).

For completeness, standard Gaussian group regularization supplies density in part (iii). If $x \in H$ and ρ_s is a centered Gaussian approximate identity, then

$$x_s = \int_{\mathbb{R}} \rho_s(t) T_c(t) x dt \longrightarrow x.$$

Integration by parts expresses $A_c^n x_s$ using $\rho_s^{(n)}$; Gaussian derivative estimates give $\|A_c^n x_s\| \leq C_s D_s^n \sqrt{n!} \|x\|$. Hence x_s is an analytic vector. Analytic vectors are quasi-analytic because Stirling's formula makes the inverse-root series dominate a constant multiple of the harmonic series. This is the familiar analytic-vector regularization mechanism [7]. $\square$

Sharpness at the constant coefficient

The next theorem records the full constant-model classification and the smooth obstruction from the preceding packet version. Let $\mathcal{F}$ be the unitary Fourier transform with $\mathcal{F}(f') = i\xi \widehat{f}$. For $A = -d/dx + 1$ and nonzero $g \in C^\infty(A)$, put

$$d\mu_g(\xi) = |\widehat{g}(\xi)|^2 d\xi, \quad r(\xi) = \frac{1}{1 - i\xi},$$

and write $\mathcal{P}$ for the complex polynomials.

Theorem 3 (Exact constant criterion and obstruction). *For $A = -d/dx + 1$ on $H^1(\mathbb{R}) \subset L^2(\mathbb{R})$:*

- (i) *The solution of $Af = g$ is Krylov if and only if $r \in \overline{\mathcal{P}}^{L^2(\mu_g)}$.*
- (ii) *If $\int e^{\varepsilon|\xi|} d\mu_g(\xi) < \infty$ for some $\varepsilon > 0$, then polynomials are dense in $L^2(\mu_g)$ and the problem is Krylov-solvable. In particular, bounded Fourier support is sufficient without any smoothness assumption on $\widehat{g}$.*
- (iii) *There is a nonzero $g \in \mathcal{S}(\mathbb{R})$ for which the solution is not Krylov. This datum is not quasi-analytic for A .*

Proof. Under $\mathcal{F}$, A is multiplication by $m(\xi) = 1 - i\xi$ and A^{-1} is multiplication by $r = m^{-1}$. The map

$$J_g : L^2(\mu_g) \rightarrow L^2(\mathbb{R}), \quad J_g p = p\hat{g},$$

is an isometry with closed range. Since polynomials in m and in ξ coincide,

$$\mathcal{F}(\overline{\mathcal{K}(A, g)}) = J_g \left(\overline{\mathcal{P}^{L^2(\mu_g)}} \right), \quad \widehat{A^{-1}g} = J_g r.$$

This proves (i).

For (ii), if $h \in L^2(\mu_g)$ is polynomial-orthogonal, the finite complex measure $d\eta = \bar{h} d\mu_g$ has an exponential moment of half the given order by Cauchy–Schwarz. Its bilateral Laplace transform is holomorphic on a strip and every derivative at zero vanishes. The identity theorem and uniqueness of Fourier transforms of finite measures give $\eta = 0$ [8, Chapter 9]. Thus the polynomial orthogonal complement is trivial.

For (iii), let ν be the lognormal probability measure

$$d\nu(x) = \frac{1}{x\sqrt{2\pi}} \exp\left(-\frac{(\log x)^2}{2}\right) dx \quad (x > 0) \quad (12)$$

and $h_0(x) = \sin(2\pi \log x)$. If $X = e^Y$ with Y standard normal, then for every integer $n \geq 0$,

$$\int_0^\infty x^n h_0(x) d\nu(x) = \operatorname{Im} \exp\left(\frac{(n + 2\pi i)^2}{2}\right) = 0. \quad (13)$$

The signed measure $\sigma = h_0\nu$ is nonzero, so its Cauchy transform

$$F(z) = \int_0^\infty \frac{1}{x - z} d\sigma(x), \quad \operatorname{Im} z < 0,$$

is not identically zero. Indeed, otherwise its Poisson convolutions would all vanish. The approximate-identity limit would then force $\sigma = 0$. Choose $z_0 = u + iv$ with $v < 0$ and $F(z_0) \neq 0$. Put

$$a = -1/v > 0, \quad b = -au, \quad T(x) = ax + b,$$

and transport ν by T . If $\mu = T_\# \nu$ and $h(Tx) = h_0(x)$, equation (13) says that h annihilates every polynomial in $L^2(\mu)$, whereas

$$\int r(\xi) h(\xi) d\mu(\xi) = \frac{i}{a} F(z_0) \neq 0,$$

because $1 - i(ax + b) = -ia(x - z_0)$. Hence $r \notin \overline{\mathcal{P}^{L^2(\mu)}}$.

Let w be the density of μ and choose $\hat{g} = \sqrt{w}$. Before the affine change of variable, the square-root density, extended by zero to the negative half-line, is a constant multiple of

$$q_0(x) = x^{-1/2} \exp\left(-\frac{(\log x)^2}{4}\right) \quad (x > 0).$$

Every derivative has the form $x^{-k-1/2} P_k(\log x) e^{-(\log x)^2/4}$. The log-Gaussian dominates every exponential as $\log x \rightarrow \pm\infty$, proving flatness at zero and Schwartz decay at infinity. Therefore $\hat{g}$, and hence g , is Schwartz and $\mu_g = \mu$. Part (i) now proves non-solvability.

Finally this example is outside the positive class of Theorem 2. With $\xi = aX + b$, for all large n and on $Y = \log X \in [2n, 2n + 1]$, one has $|\xi| \geq aX/2$ and

$$\int_{2n}^{2n+1} e^{2ny-y^2/2} \frac{dy}{\sqrt{2\pi}} \geq c_0 e^{2n^2}.$$

Consequently $\|A^n g\| \geq c_1 (a/2)^n e^{n^2}$ for all large n , so $\sum_n \|A^n g\|^{-1/n} < \infty$. The Schwartz datum is A -smooth but not A -quasi-analytic. $\square$

Verification notes

The following points received separate adversarial checks.

- The integrating factor is used as a unitary map from the weighted space, not incorrectly as a bounded similarity on unweighted L^2 .
- The translation estimate has the correct orientation: right translation compares $\omega(y - t)$ with $\omega(y)$ and decays for positive time because $\operatorname{Re} c \geq \delta$.
- The Denjoy–Carleman argument is applied to scalar group orbits on every compact real interval; the group, rather than only a semigroup, removes an endpoint uniqueness issue.
- The maximal-domain statement follows by unitary definition and the weak product rule. The familiar H^1 domain is asserted only when c is bounded.
- The negative example exhibits both a polynomial-orthogonal witness and a nonzero pairing with the specific resolvent; moment indeterminacy alone is not used.
- The lower power-norm bound confirms that the counterexample does not contradict the quasi-analytic theorem.

The accompanying script checks the lognormal moment formula at high precision and evaluates one sample Cauchy pairing as a regression test. These numerical checks are not used in the proof. The detailed report is `verification.md`.

Novelty check and limitations

The run indexes and a bounded arXiv/web search were checked using the exact source title and authors, “Krylov solvability Friedrichs”, “ $-d/dx + c(x)$ Krylov”, “analytic vector Krylov-solvable”, stable semigroups, and quasi-analytic vectors. The closest works inspected were arXiv:1811.08202, 2001.08127, 2102.13626, and 2210.04752 [3, 2, 4, 5]. They contain the general framework, self-adjoint/skew-adjoint mechanisms, perturbation theory, and bounded or compact normal examples. No later paper explicitly resolving Question 3, and no statement of Theorem 1 in this Krylov form, was located in this bounded search. This is not proof of novelty.

The stable-group theorem is a short application of the classical Denjoy–Carleman theorem and may be folklore. The candidate-new contribution is its coefficient-wide Friedrichs application together with the exact weighted model and the quasi-analytic-versus-smooth separation. The coefficient may have unbounded imaginary part under the stated local regularity, but the realization is then the maximal domain rather than necessarily H^1 . Matrix-valued Friedrichs systems and a simpler intrinsic classification of every smooth datum remain outside the packet.

Human-review recommendation

Preserve as a coefficient-wide strong partial-resolution candidate. A reviewer should focus on the precise Denjoy–Carleman criterion used after (4), the maximal-domain identification under only $c \in L^2_{\text{loc}}$, and the density estimate in Gaussian group regularization. If the natural interpretation of the open-ended source question is “give useful sharp conditions across the scalar class”, this packet can reasonably be viewed as a full scalar resolution. Protocol-wise, **partial** remains the safer label pending that judgment.

References

- [1] N. A. Caruso and A. Michelangeli, *Open problems and perspectives on solving Friedrichs systems by Krylov approximation*, arXiv:2406.05777v2 (2024).
- [2] N. A. Caruso and A. Michelangeli, *Krylov solvability of unbounded inverse linear problems*, Integral Equations Operator Theory 93 (2021), Paper No. 1; arXiv:2001.08127.
- [3] N. A. Caruso, A. Michelangeli, and P. Novati, *On Krylov solutions to infinite-dimensional inverse linear problems*, Calcolo 56 (2019), Paper No. 32; arXiv:1811.08202.
- [4] N. A. Caruso and A. Michelangeli, *Krylov solvability under perturbations of abstract inverse linear problems*, Journal of Applied Analysis 29 (2023), 3–29; arXiv:2102.13626.
- [5] N. A. Caruso, *A note on the Krylov solvability of compact normal operators on Hilbert space*, Complex Analysis and Operator Theory 17 (2023); arXiv:2210.04752.
- [6] P. R. Chernoff, *Quasi-analytic vectors and quasi-analytic functions*, Bulletin of the American Mathematical Society 81 (1975), 637–646.
- [7] E. Nelson, *Analytic vectors*, Annals of Mathematics (2) 70 (1959), 572–615.
- [8] W. Rudin, *Real and Complex Analysis*, 3rd ed., McGraw–Hill, 1987.